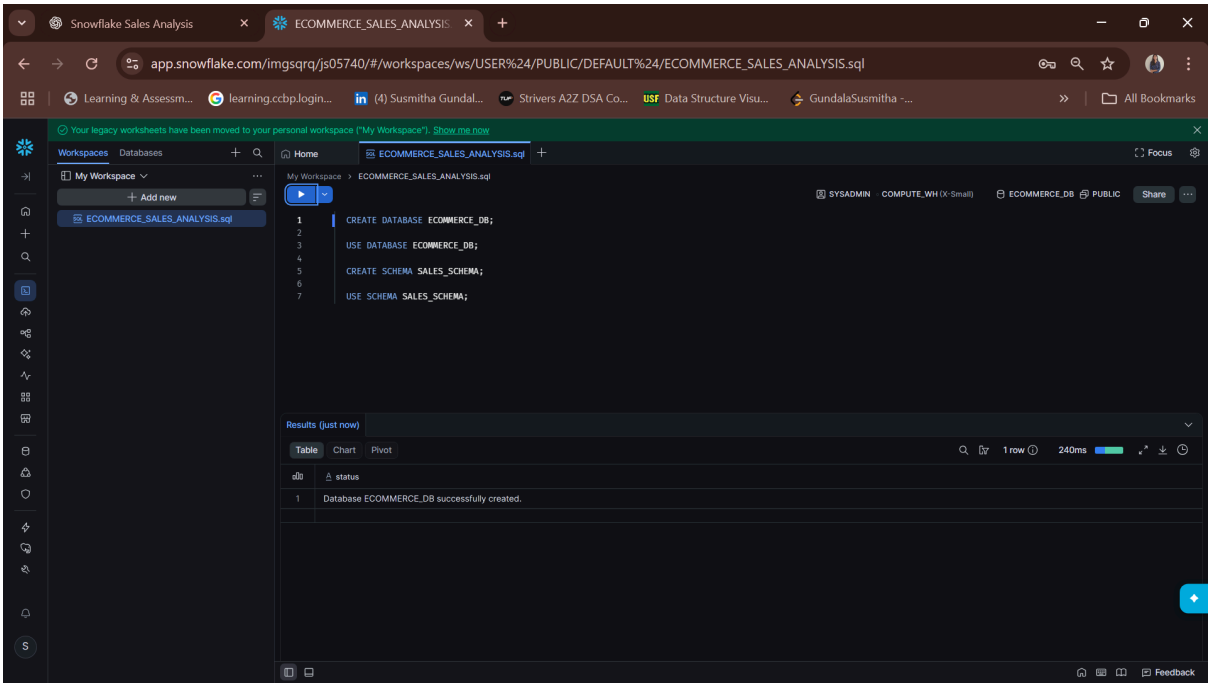
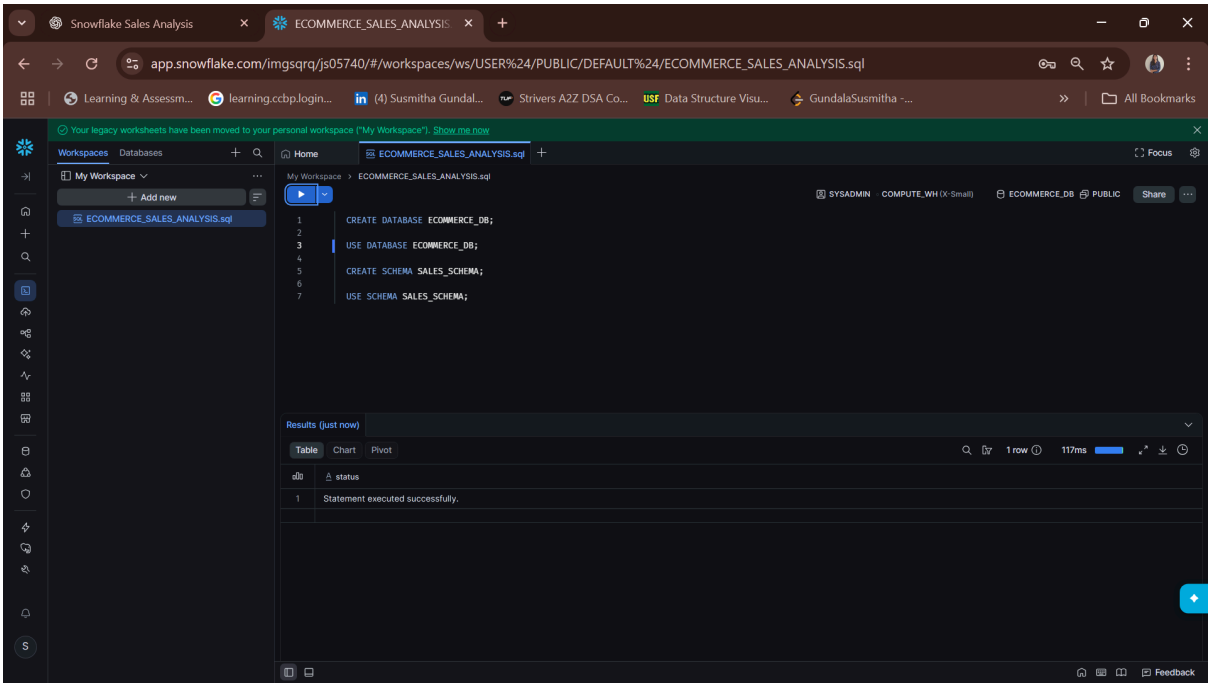
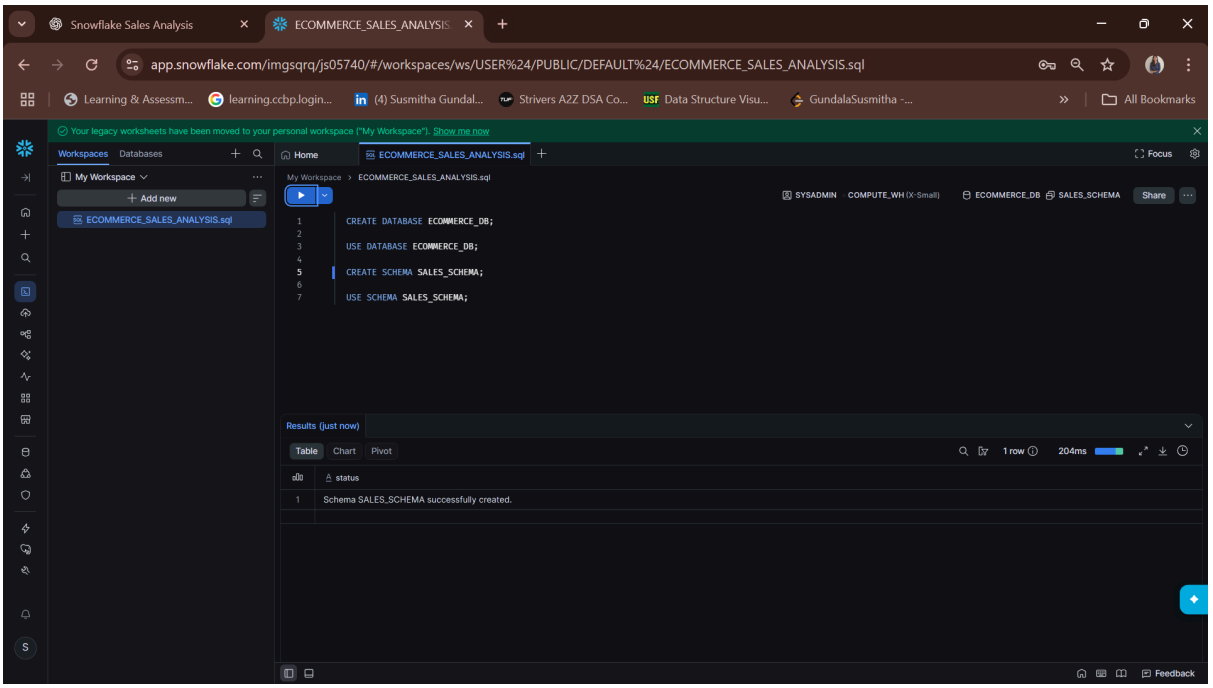
Snowflake-Based E-commerce Sales Analysis: Database Creation, CSV Data Loading, and SQL Business Analytics

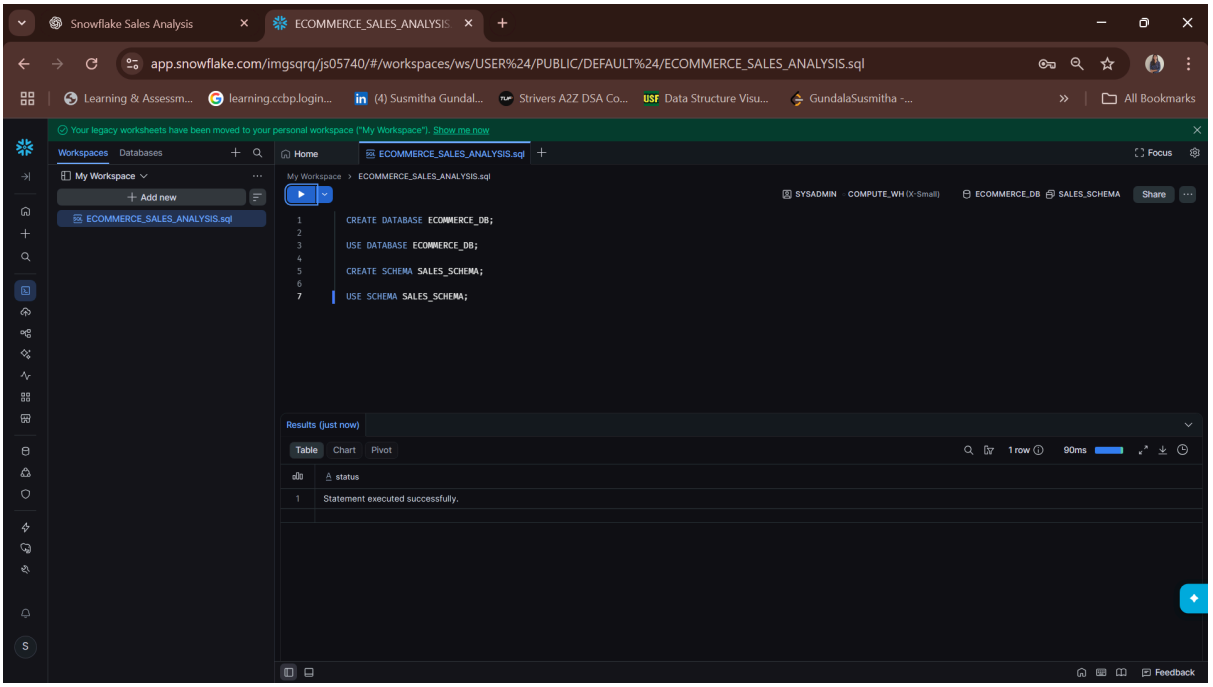
Database and Schema Creation

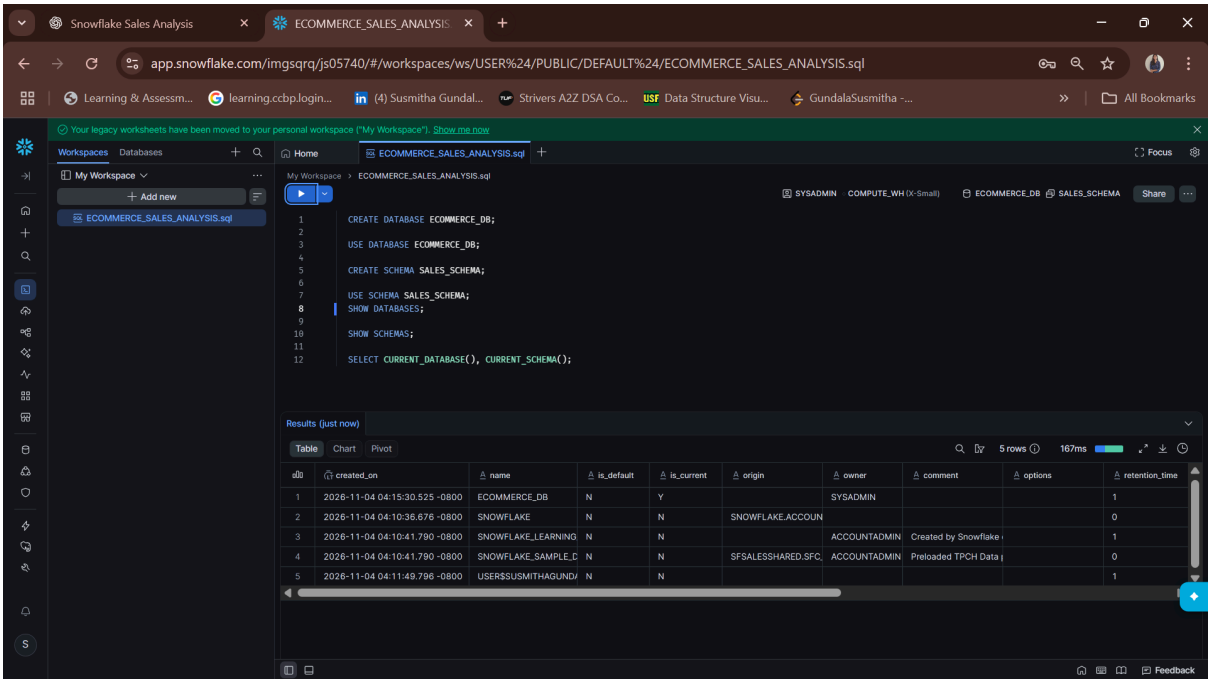
```sql
CREATE DATABASE ECOMMERCE_DB;
USE DATABASE ECOMMERCE_DB;
CREATE SCHEMA SALES_SCHEMA;
USE SCHEMA SALES_SCHEMA;
SHOW DATABASES;
SHOW SCHEMAS;
```

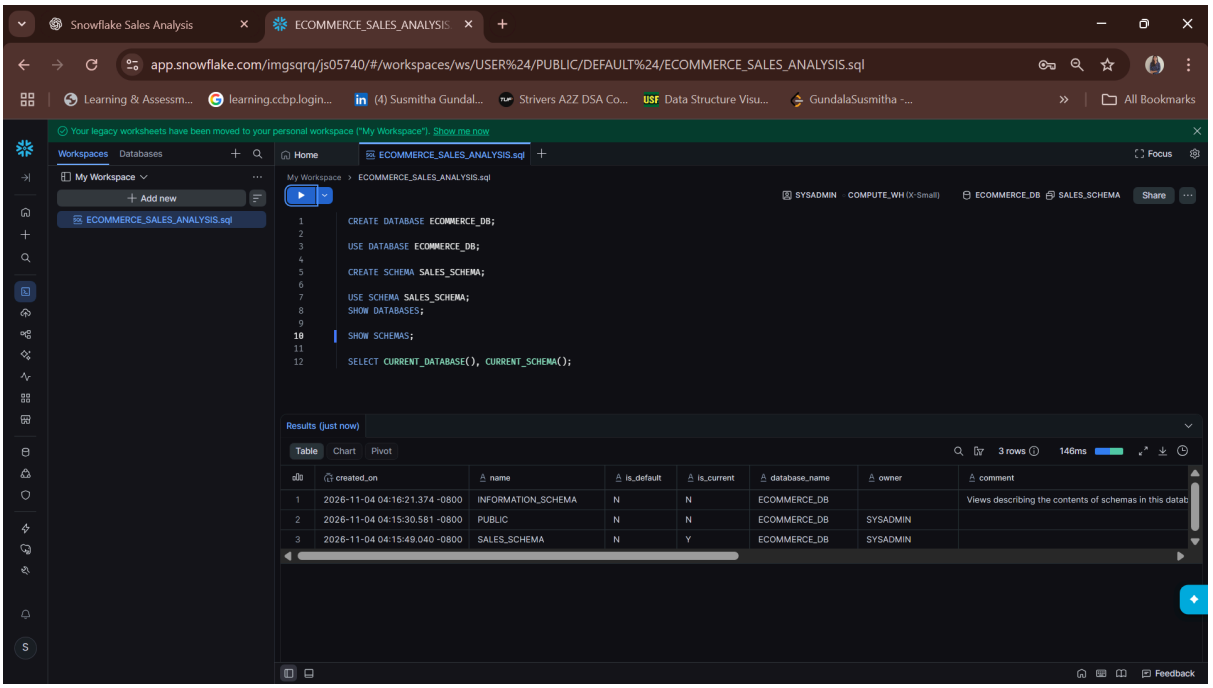



Snowflake Sales Analysis
ECOMMERCE_SALES_ANALYSIS
app.snowflake.com/imgsqrq/js05740/#/workspaces/ws/USER%24/PUBLIC/DEFAULT%24/ECOMMERCE_SALES_ANALYSIS.sql
Your legacy worksheets have been moved to your personal workspace ("My Workspace"). Show me now
Workspaces
Databases
My Workspace
Add new
ECOMMERCE_SALES_ANALYSIS.sql
Home
ECOMMERCE_SALES_ANALYSIS.sql
Focus
My Workspace › ECOMMERCE_SALES_ANALYSIS.sql
SYSADMIN · COMPUTE_WH (X-Small)
ECOMMERCE_DB PUBLIC
Share
1 CREATE DATABASE ECOMMERCE_DB;
2
3 USE DATABASE ECOMMERCE_DB;
4
5 CREATE SCHEMA SALES_SCHEMA;
6
7 USE SCHEMA SALES_SCHEMA;
Results (just now)
Table
Chart
Pivot
1 row
117ms
status
1 Statement executed successfully.
Feedback


Snowflake Sales Analysis
ECOMMERCE_SALES_ANALYSIS
app.snowflake.com/imgsqrq/js05740/#/workspaces/ws/USER%24/PUBLIC/DEFAULT%24/ECOMMERCE_SALES_ANALYSIS.sql
Your legacy worksheets have been moved to your personal workspace ("My Workspace"). Show me now
Workspaces
Databases
My Workspace
Add new
ECOMMERCE_SALES_ANALYSIS.sql
Home
ECOMMERCE_SALES_ANALYSIS.sql
Focus
My Workspace › ECOMMERCE_SALES_ANALYSIS.sql
SYSADMIN · COMPUTE_WH (X-Small)
ECOMMERCE_DB SALES_SCHEMA
Share
1 CREATE DATABASE ECOMMERCE_DB;
2
3 USE DATABASE ECOMMERCE_DB;
4
5 CREATE SCHEMA SALES_SCHEMA;
6
7 USE SCHEMA SALES_SCHEMA;
Results (just now)
Table
Chart
Pivot
1 row
204ms
status
1 Schema SALES_SCHEMA successfully created.
Feedback


Snowflake Sales Analysis
ECOMMERCE_SALES_ANALYSIS
app.snowflake.com/imgsqrq/js05740/#/workspaces/ws/USER%24/PUBLIC/DEFAULT%24/ECOMMERCE_SALES_ANALYSIS.sql
Learning & Assessm...
learning.ccbp.login...
(4) Susmitha Gundal...
Strivers A2Z DSA Co...
Data Structure Visu...
GundalaSusmitha -...
All Bookmarks
Your legacy worksheets have been moved to your personal workspace ("My Workspace"). Show me now
Workspaces
Databases
My Workspace
Add new
ECOMMERCE_SALES_ANALYSIS.sql
Home
ECOMMERCE_SALES_ANALYSIS.sql
Focus
My Workspace > ECOMMERCE_SALES_ANALYSIS.sql
SYSADMIN
COMPUTE_WH (X-Small)
ECOMMERCE_DB
SALES_SCHEMA
Share
1 CREATE DATABASE ECOMMERCE_DB;
2
3 USE DATABASE ECOMMERCE_DB;
4
5 CREATE SCHEMA SALES_SCHEMA;
6
7 USE SCHEMA SALES_SCHEMA;
Results (just now)
Table
Chart
Pivot
1 row
90ms
status
1
Statement executed successfully.
Feedback



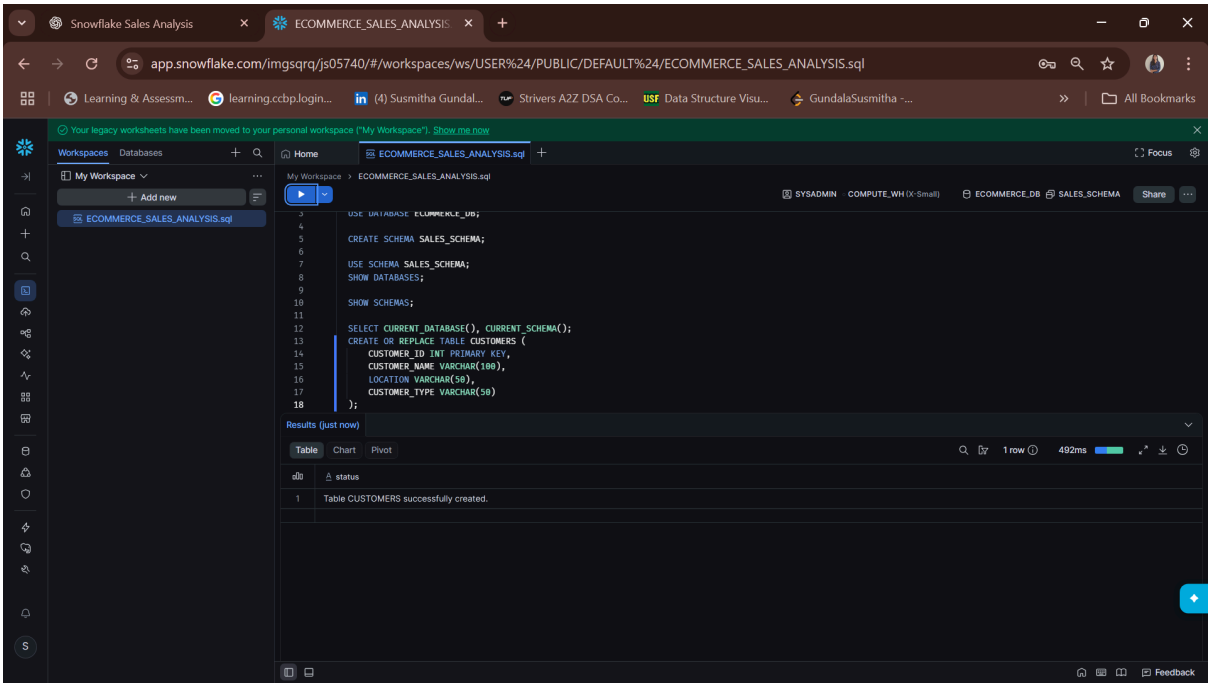


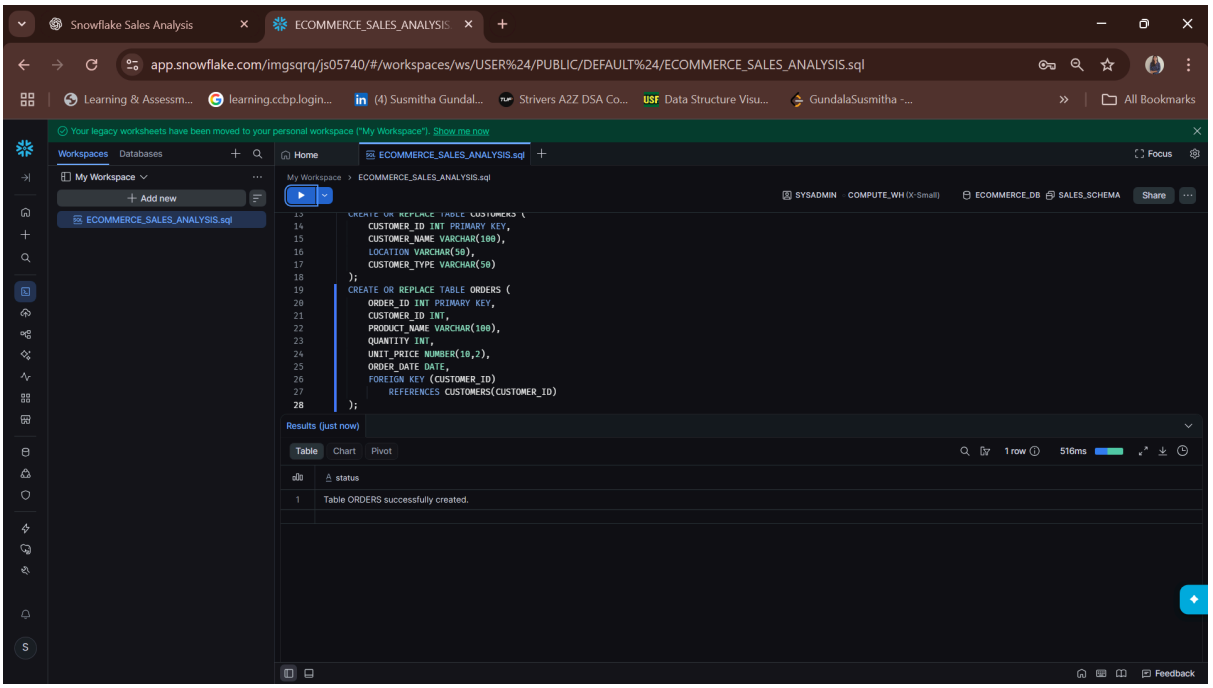
Relational Table Creation

```sql
SELECT CURRENT_DATABASE(), CURRENT_SCHEMA();

CREATE OR REPLACE TABLE CUSTOMERS (
    CUSTOMER_ID INT PRIMARY KEY,
    CUSTOMER_NAME VARCHAR(100),
    LOCATION VARCHAR(50),
    CUSTOMER_TYPE VARCHAR(50)
);
```

```sql
CREATE OR REPLACE TABLE ORDERS (
    ORDER_ID INT PRIMARY KEY,
    CUSTOMER_ID INT,
    PRODUCT_NAME VARCHAR(100),
    QUANTITY INT,
    UNIT_PRICE NUMBER(10,2),
    ORDER_DATE DATE,
    FOREIGN KEY (CUSTOMER_ID)
        REFERENCES CUSTOMERS(CUSTOMER_ID)
);
```



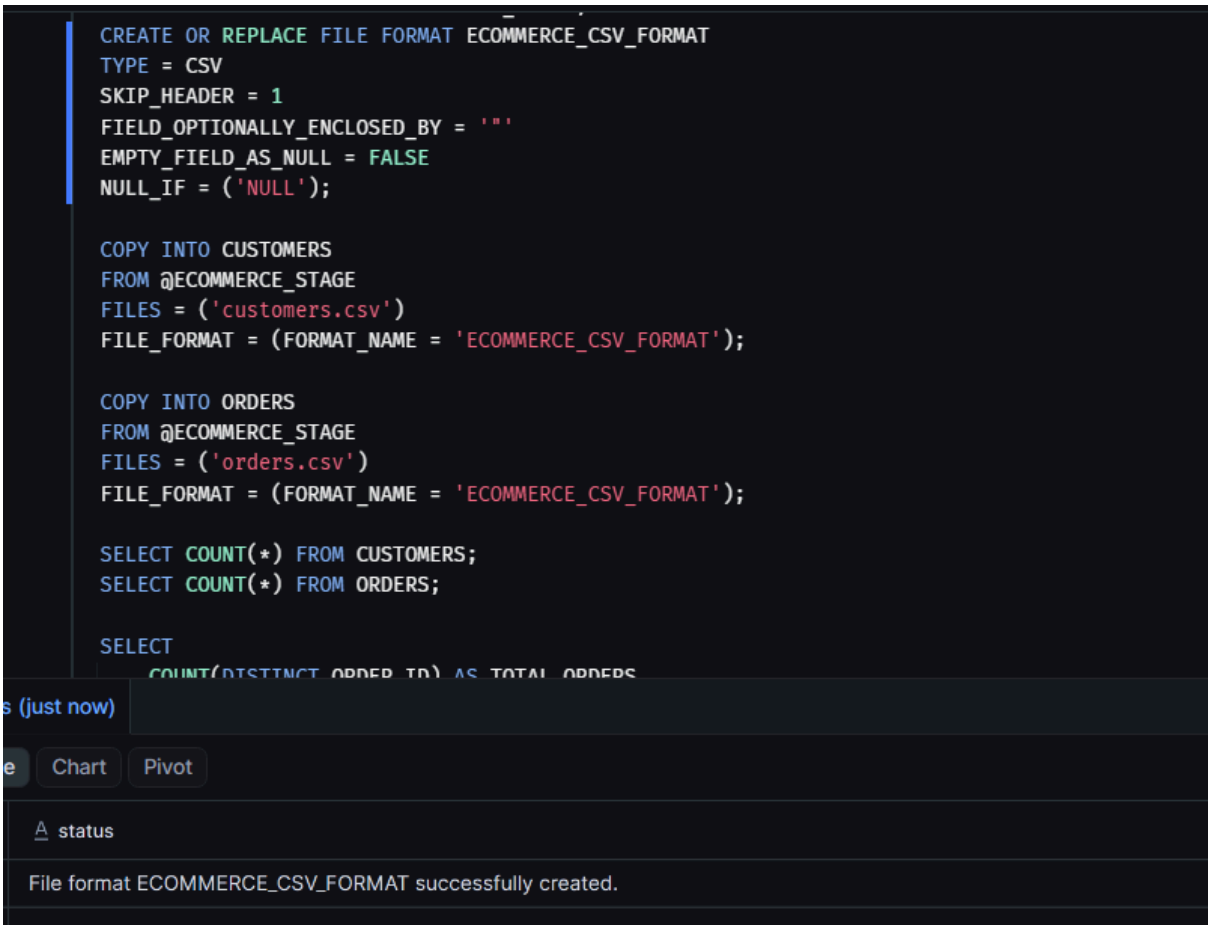


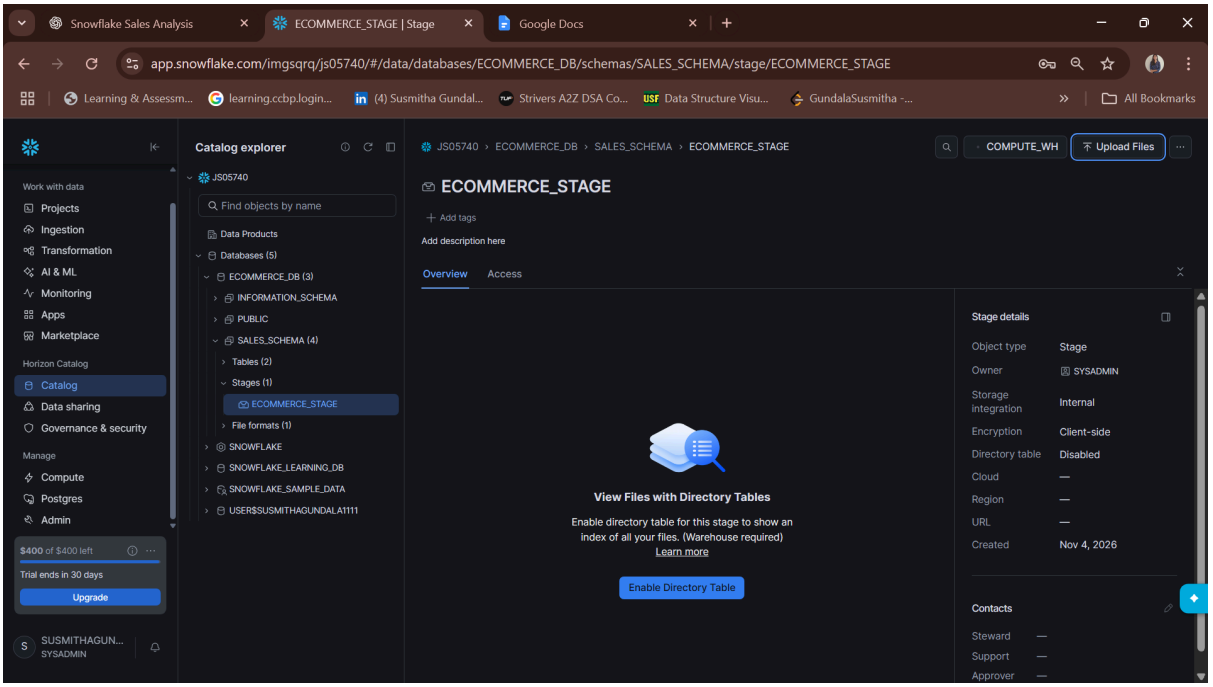
Stage Creation and CSV Upload , Create CSV file format , Upload CSV files and Loading Data into Snowflake Tables

```sql
CREATE OR REPLACE STAGE ECOMMERCE_STAGE;
CREATE OR REPLACE FILE FORMAT ECOMMERCE_CSV_FORMAT
TYPE = CSV
SKIP_HEADER = 1
FIELD_OPTIONALLY_ENCLOSED_BY = '"'
EMPTY_FIELD_AS_NULL = FALSE
NULL_IF = ('NULL');
```

```
COPY INTO CUSTOMERS
FROM @ECOMMERCE_STAGE
FILES = ('customers.csv')
FILE_FORMAT = (FORMAT_NAME = 'ECOMMERCE_CSV_FORMAT');

COPY INTO ORDERS
FROM @ECOMMERCE_STAGE
FILES = ('orders.csv')
FILE_FORMAT = (FORMAT_NAME = 'ECOMMERCE_CSV_FORMAT');
```





SQL Query Implementations

```sql
INSERT INTO CUSTOMERS VALUES
(101, 'Ananya', 'Chennai', 'Retail'),
(102, 'Rahul', 'Bengaluru', 'Wholesale'),
(103, 'Meera', 'Hyderabad', 'Retail'),
(104, 'Arjun', 'Mumbai', 'Corporate'),
(105, 'Divya', 'Chennai', 'Wholesale'),
(106, 'Kiran', 'Pune', 'Retail'),
(107, 'Sneha', 'Hyderabad', 'Corporate'),
(108, 'Vikram', 'Bengaluru', 'Retail'),
(109, 'Priya', 'Mumbai', ' '),
(110, 'Nikhil', 'Chennai', NULL);

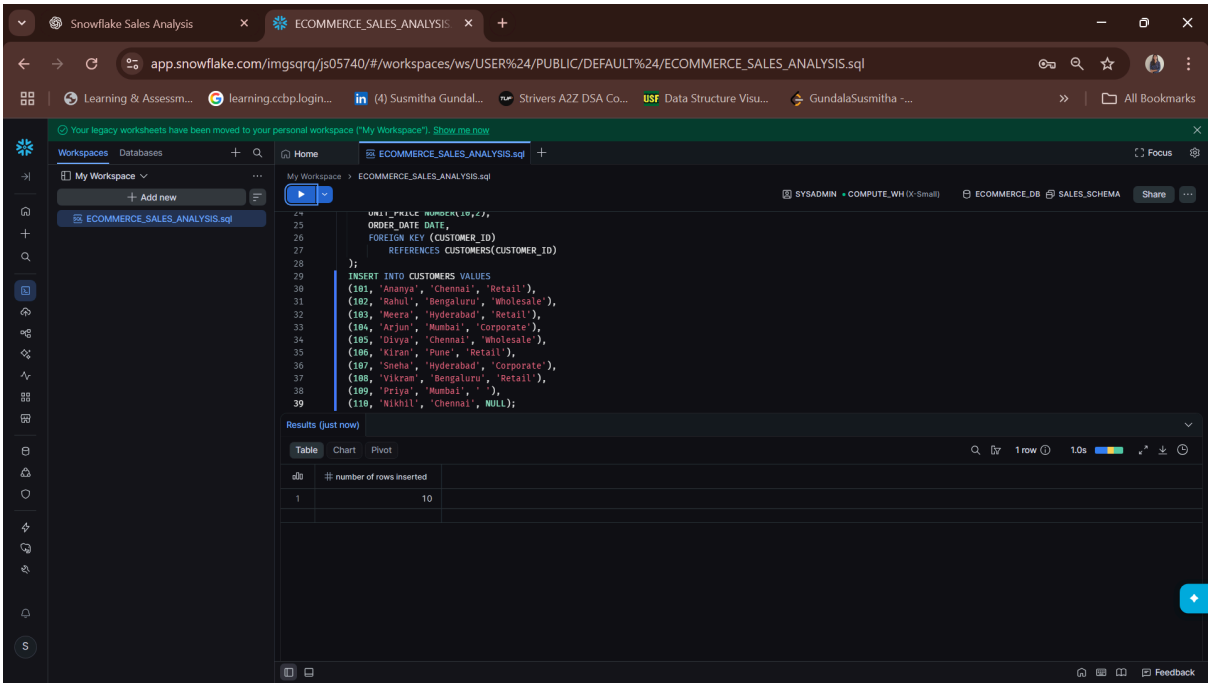
INSERT INTO ORDERS VALUES
(5001, 101, 'Laptop', 1, 55000, '2026-10-01'),
(5002, 102, 'Mouse', 2, 750, '2026-10-02'),
(5003, 103, 'Keyboard', 1, 1800, '2026-10-02'),
(5004, 104, 'Monitor', 2, 12000, '2026-10-03'),
(5005, 105, 'Laptop', 1, 55000, '2026-10-04'),
(5006, 106, 'Mouse', 3, 750, '2026-10-04'),
(5007, 107, 'Headphones', 2, 2500, '2026-10-05'),
(5008, 108, 'Keyboard', 2, 1800, '2026-10-06'),
(5009, 109, 'Monitor', 1, 12000, '2026-10-06'),
(5010, 110, 'Laptop', 1, 55000, '2026-10-07'),
(5011, 101, 'Headphones', 1, 2500, '2026-10-08'),
(5012, 102, 'Mouse', 1, 750, '2026-10-08'),
(5013, 103, 'Monitor', 1, 12000, '2026-10-09'),
(5014, 104, 'Keyboard', 1, 1800, '2026-10-10'),
```

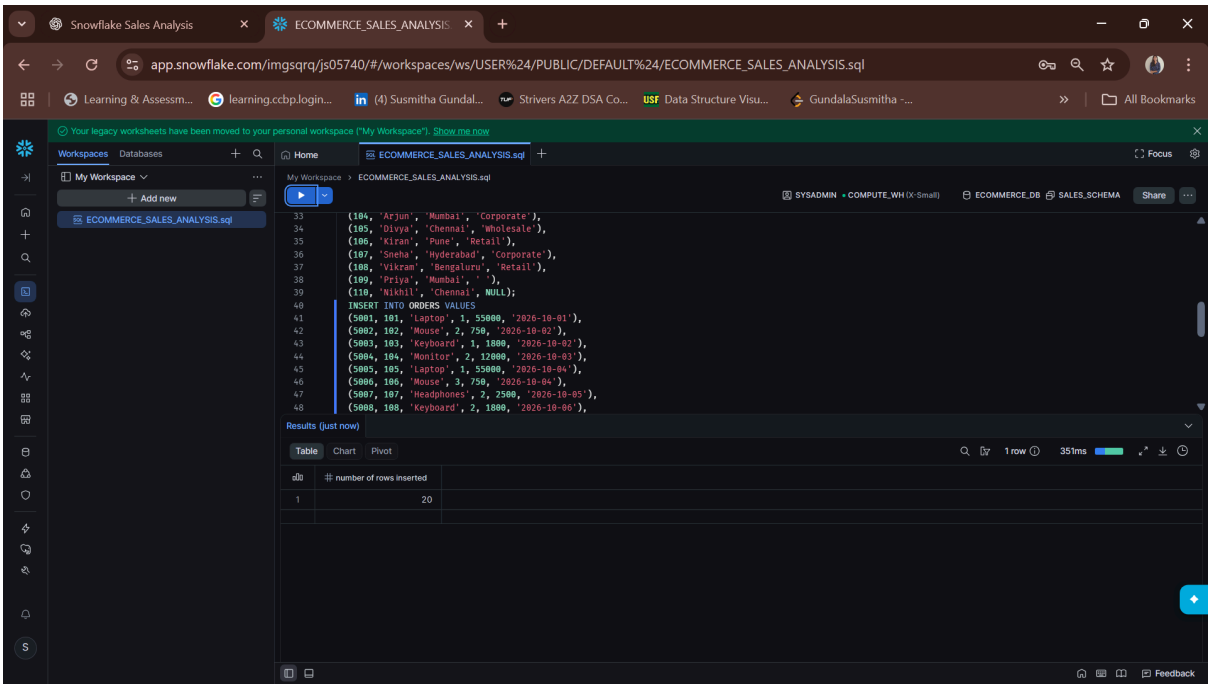
```
(5015, 105, 'Headphones', 2, 2500, '2026-10-10'),
(5016, 106, 'Laptop', 1, 55000, '2026-10-11'),
(5017, 107, 'Mouse', 2, 750, '2026-10-12'),
(5018, 108, 'Monitor', 1, 12000, '2026-10-12'),
(5019, 109, 'Keyboard', 2, 1800, '2026-10-13'),
(5020, 110, 'Headphones', 1, 2500, '2026-10-14');

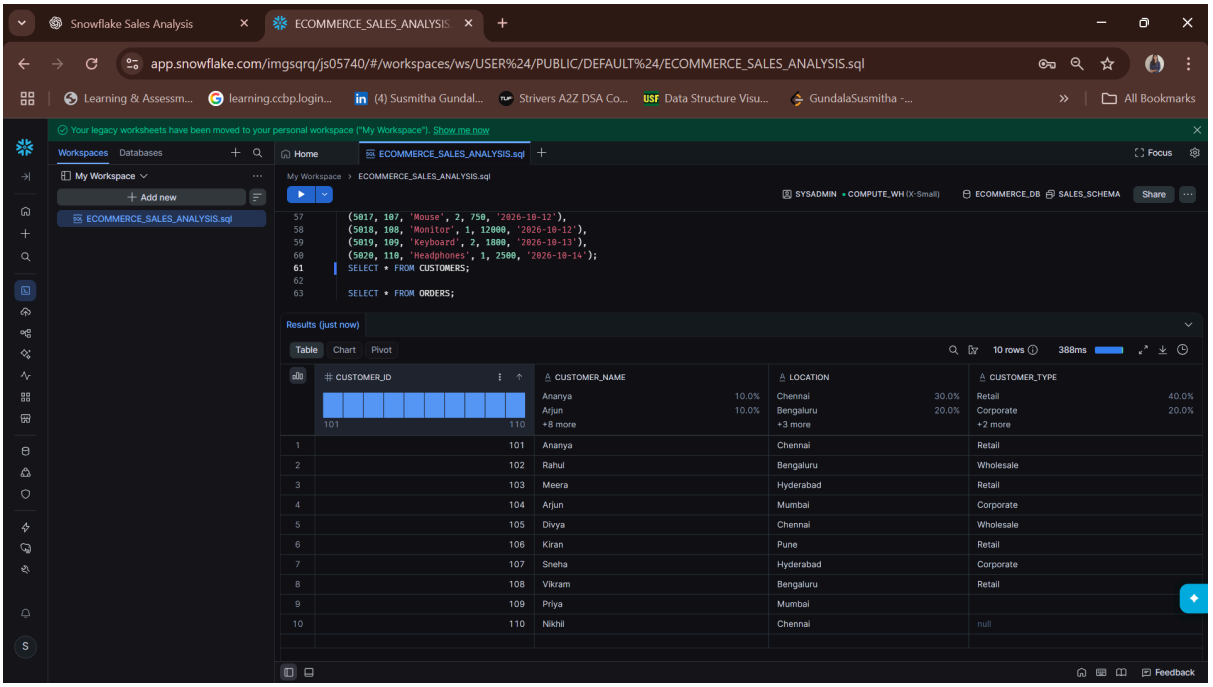
SELECT * FROM CUSTOMERS;

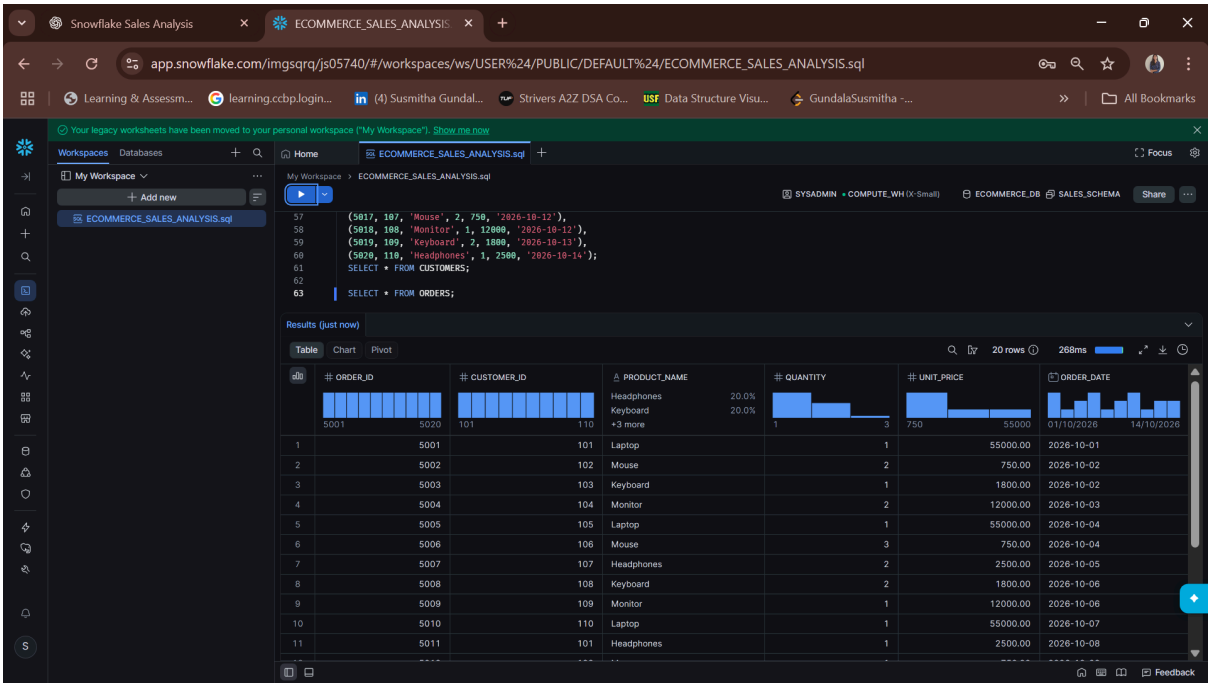
SELECT * FROM ORDERS;

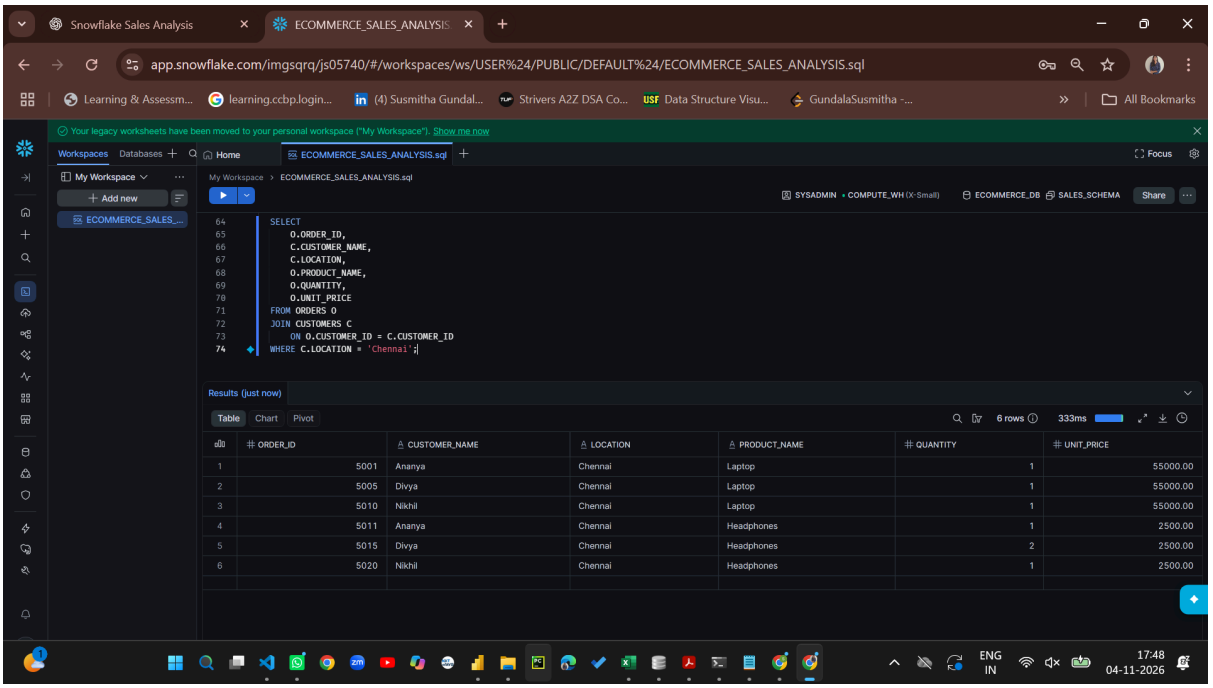
SELECT
    O.ORDER_ID,
    C.CUSTOMER_NAME,
    C.LOCATION,
    O.PRODUCT_NAME,
    O.QUANTITY,
    O.UNIT_PRICE
FROM ORDERS O
JOIN CUSTOMERS C
    ON O.CUSTOMER_ID = C.CUSTOMER_ID
WHERE C.LOCATION = 'Chennai'
```


Snowflake Sales Analysis
ECOMMERCE_SALES_ANALYSIS
app.snowflake.com/imgsqrq/js05740/#/workspaces/ws/USER%24/PUBLIC/DEFAULT%24/ECOMMERCE_SALES_ANALYSIS.sql
Your legacy worksheets have been moved to your personal workspace ("My Workspace"). Show me now
Workspaces
Databases
My Workspace
Add new
ECOMMERCE_SALES_ANALYSIS.sql
Home
Focus
SYSADMIN • COMPUTE_WH (X-Small)
ECOMMERCE_DB
SALES_SCHEMA
Share
ORDER_DATE DATE,
FOREIGN KEY (CUSTOMER_ID)
REFERENCES CUSTOMERS(CUSTOMER_ID)
);
INSERT INTO CUSTOMERS VALUES
(101, 'Ananya', 'Chennai', 'Retail'),
(102, 'Rahul', 'Bengaluru', 'Wholesale'),
(103, 'Meera', 'Hyderabad', 'Retail'),
(104, 'Arjun', 'Mumbai', 'Corporate'),
(105, 'Divya', 'Chennai', 'Wholesale'),
(106, 'Kiran', 'Pune', 'Retail'),
(107, 'Sneha', 'Hyderabad', 'Corporate'),
(108, 'Vikram', 'Bengaluru', 'Retail'),
(109, 'Priya', 'Mumbai', ' '),
(110, 'Nikhil', 'Chennai', NULL);
Results (just now)
Table
Chart
Pivot
1 row
1.0s
number of rows inserted
1
10
Feedback


Snowflake Sales Analysis
ECOMMERCE_SALES_ANALYSIS
app.snowflake.com/imgsqrq/js05740/#/workspaces/ws/USER%24/PUBLIC/DEFAULT%24/ECOMMERCE_SALES_ANALYSIS.sql
Your legacy worksheets have been moved to your personal workspace ("My Workspace"). Show me now
Workspaces
Databases
My Workspace
Add new
ECOMMERCE_SALES_ANALYSIS.sql
Home
Focus
SYSADMIN • COMPUTE_WH (X-Small)
ECOMMERCE_DB
SALES_SCHEMA
Share
(104, 'Arjun', 'Mumbai', 'Corporate'),
(105, 'Divya', 'Chennai', 'Wholesale'),
(106, 'Kiran', 'Pune', 'Retail'),
(107, 'Sneha', 'Hyderabad', 'Corporate'),
(108, 'Vikram', 'Bengaluru', 'Retail'),
(109, 'Priya', 'Mumbai', ' '),
(110, 'Nikhil', 'Chennai', NULL);
INSERT INTO ORDERS VALUES
(5001, 101, 'Laptop', 1, 55000, '2026-10-01'),
(5002, 102, 'Mouse', 2, 750, '2026-10-02'),
(5003, 103, 'Keyboard', 1, 1800, '2026-10-02'),
(5004, 104, 'Monitor', 2, 12000, '2026-10-03'),
(5005, 105, 'Laptop', 1, 55000, '2026-10-04'),
(5006, 106, 'Mouse', 3, 750, '2026-10-04'),
(5007, 107, 'Headphones', 2, 2500, '2026-10-05'),
(5008, 108, 'Keyboard', 2, 1800, '2026-10-06'),
Results (just now)
Table
Chart
Pivot
1 row
351ms
number of rows inserted
1
20
Feedback


Snowflake Sales Analysis
ECOMMERCE_SALES_ANALYSIS
app.snowflake.com/imgsqrq/js05740/#/workspaces/ws/USER%24/PUBLIC/DEFAULT%24/ECOMMERCE_SALES_ANALYSIS.sql
Your legacy worksheets have been moved to your personal workspace ("My Workspace"). Show me now
Workspaces
Databases
My Workspace
Add new
ECOMMERCE_SALES_ANALYSIS.sql
Home
ECOMMERCE_SALES_ANALYSIS.sql
Focus
My Workspace > ECOMMERCE_SALES_ANALYSIS.sql
SYSADMIN • COMPUTE_WH (X-Small)
ECOMMERCE_DB SALES_SCHEMA
Share
(5017, 107, 'Mouse', 2, 750, '2026-10-12'),
(5018, 108, 'Monitor', 1, 12000, '2026-10-12'),
(5019, 109, 'Keyboard', 2, 1800, '2026-10-13'),
(5020, 110, 'Headphones', 1, 2500, '2026-10-14');
SELECT * FROM CUSTOMERS;
SELECT * FROM ORDERS;
Results (just now)
Table
Chart
Pivot
10 rows
388ms
CUSTOMER_ID
CUSTOMER_NAME
LOCATION
CUSTOMER_TYPE
Ananya 10.0%
Arjun 10.0%
+8 more
Chennai 30.0%
Bengaluru 20.0%
+3 more
Retail 40.0%
Corporate 20.0%
+2 more
101
110
1 101 Ananya Chennai Retail
2 102 Rahul Bengaluru Wholesale
3 103 Meera Hyderabad Retail
4 104 Arjun Mumbai Corporate
5 105 Divya Chennai Wholesale
6 106 Kiran Pune Retail
7 107 Sneha Hyderabad Corporate
8 108 Vikram Bengaluru Retail
9 109 Priya Mumbai
10 110 Nikhil Chennai null
Feedback



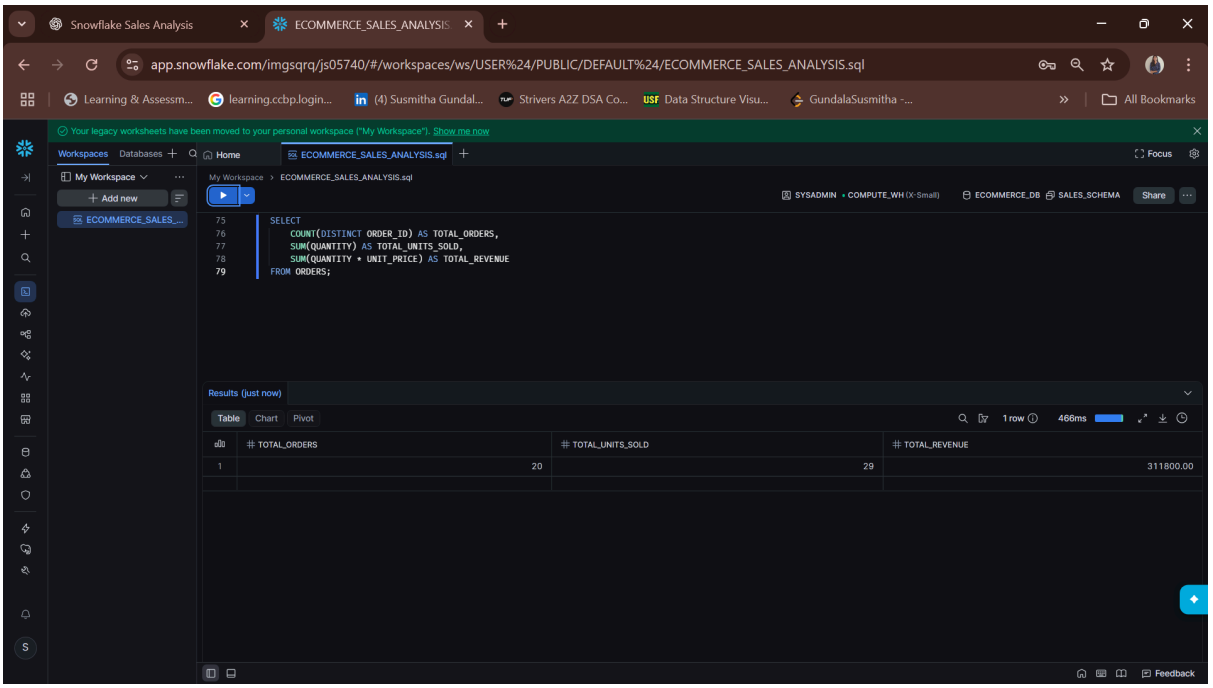


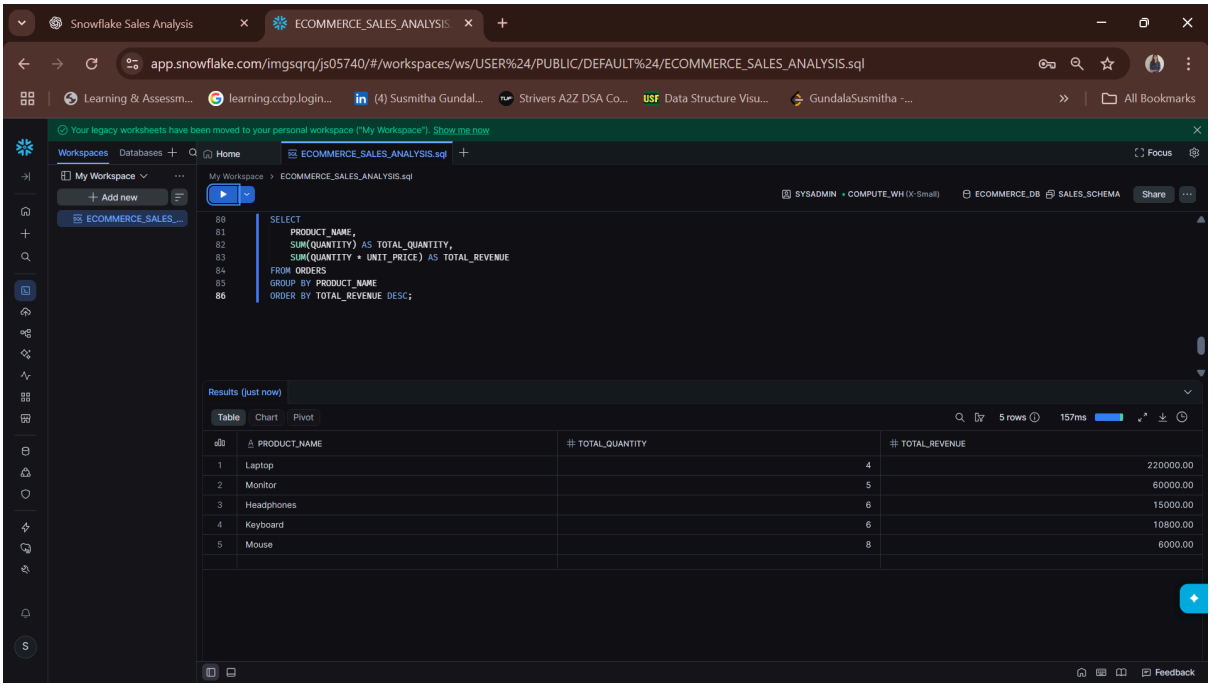

```sql
SELECT
    COUNT(DISTINCT ORDER_ID) AS TOTAL_ORDERS,
    SUM(QUANTITY) AS TOTAL_UNITS_SOLD,
    SUM(QUANTITY * UNIT_PRICE) AS TOTAL_REVENUE
FROM ORDERS;

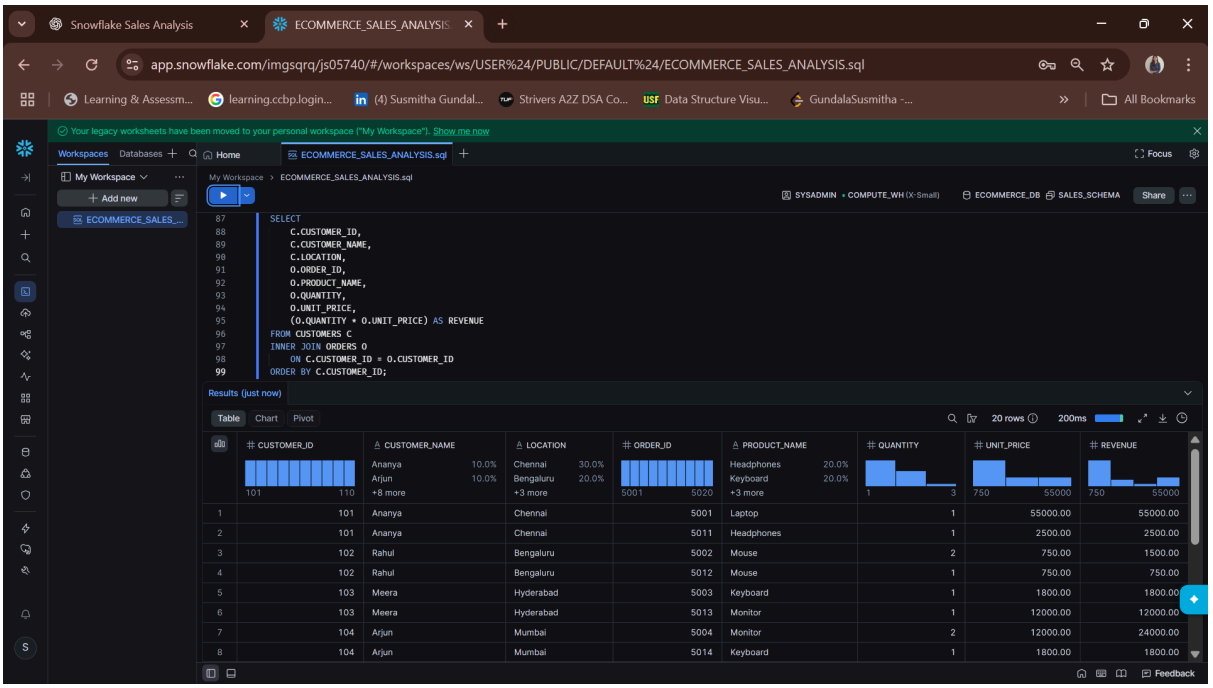
SELECT
    PRODUCT_NAME,
    SUM(QUANTITY) AS TOTAL_QUANTITY,
```

```sql
    SUM(QUANTITY * UNIT_PRICE) AS TOTAL_REVENUE
FROM ORDERS
GROUP BY PRODUCT_NAME
ORDER BY TOTAL_REVENUE DESC;

SELECT
    C.CUSTOMER_ID,
    C.CUSTOMER_NAME,
    C.LOCATION,
    O.ORDER_ID,
    O.PRODUCT_NAME,
    O.QUANTITY,
    O.UNIT_PRICE,
    (O.QUANTITY * O.UNIT_PRICE) AS REVENUE
FROM CUSTOMERS C
INNER JOIN ORDERS O
    ON C.CUSTOMER_ID = O.CUSTOMER_ID
ORDER BY C.CUSTOMER_ID;
```



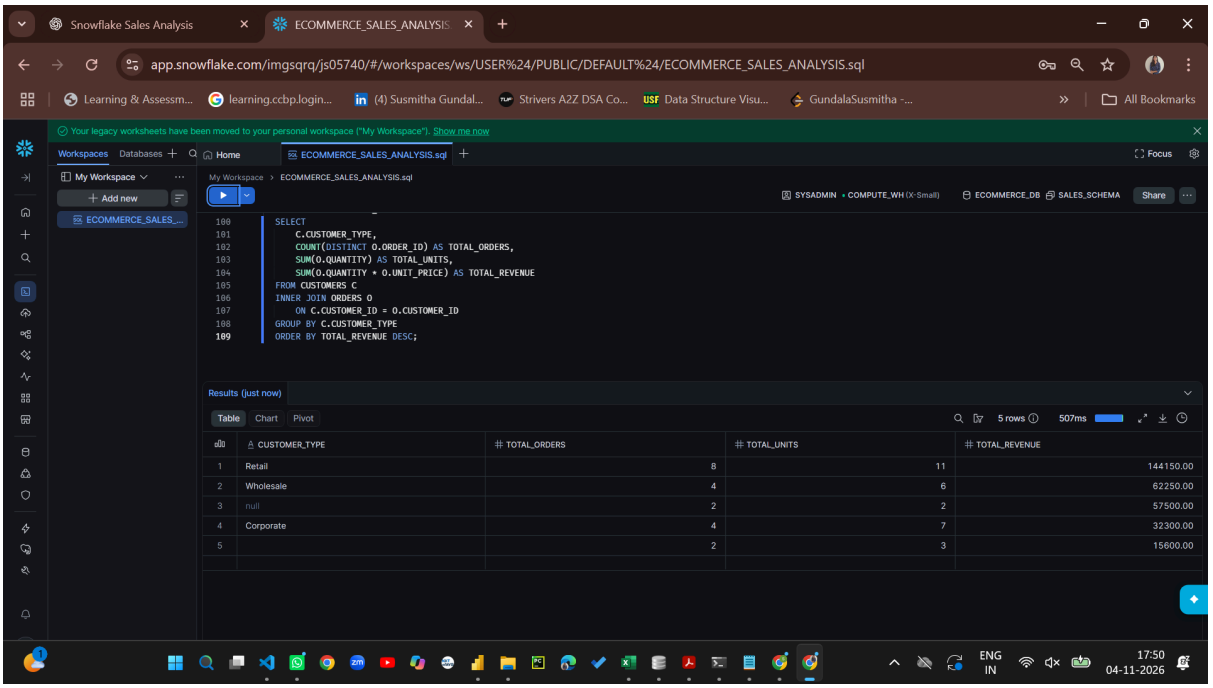


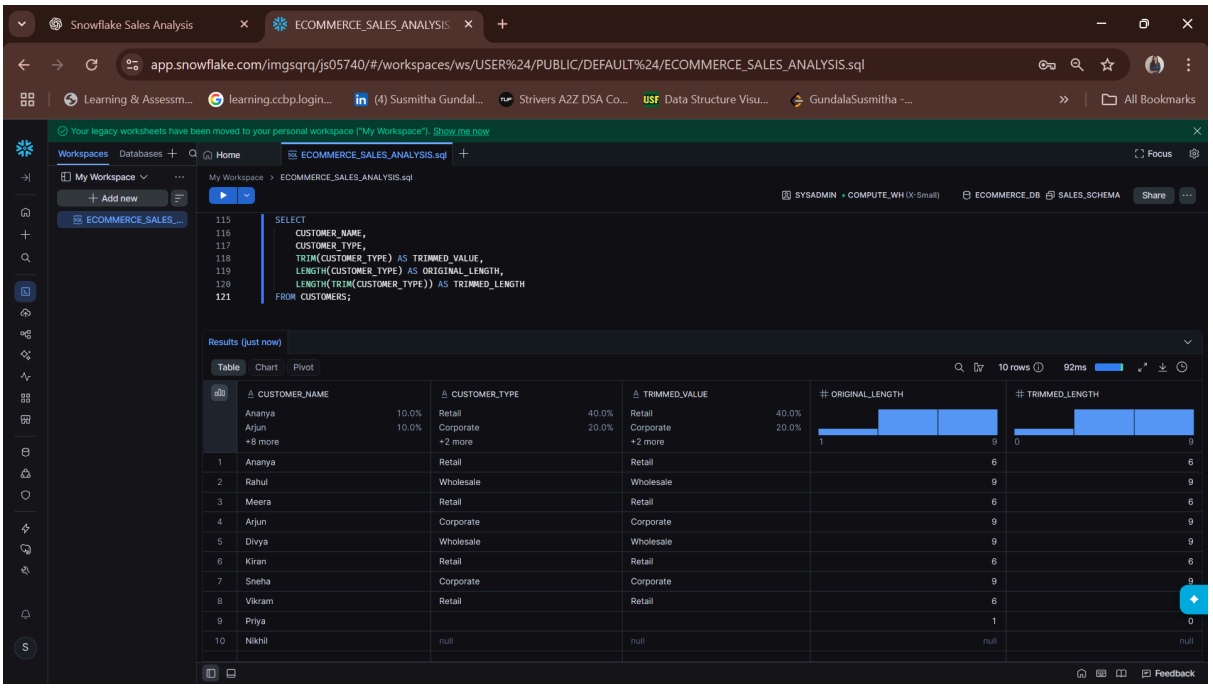



```sql
SELECT
    C.CUSTOMER_TYPE,
    COUNT(DISTINCT O.ORDER_ID) AS TOTAL_ORDERS,
    SUM(O.QUANTITY) AS TOTAL_UNITS,
    SUM(O.QUANTITY * O.UNIT_PRICE) AS TOTAL_REVENUE
FROM CUSTOMERS C
INNER JOIN ORDERS O
    ON C.CUSTOMER_ID = O.CUSTOMER_ID
GROUP BY C.CUSTOMER_TYPE
```

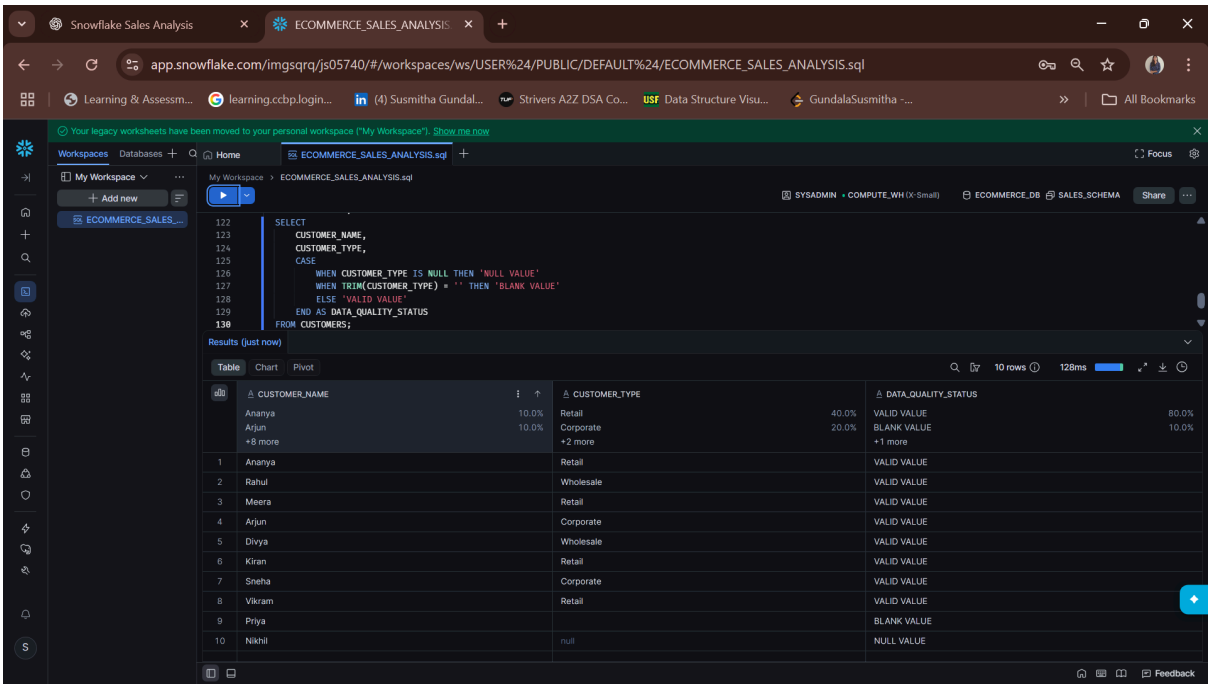
```sql
ORDER BY TOTAL_REVENUE DESC;

SELECT
    CUSTOMER_NAME,
    CUSTOMER_TYPE,
    TRIM(CUSTOMER_TYPE) AS TRIMMED_VALUE,
    LENGTH(CUSTOMER_TYPE) AS ORIGINAL_LENGTH,
    LENGTH(TRIM(CUSTOMER_TYPE)) AS TRIMMED_LENGTH
FROM CUSTOMERS;
```





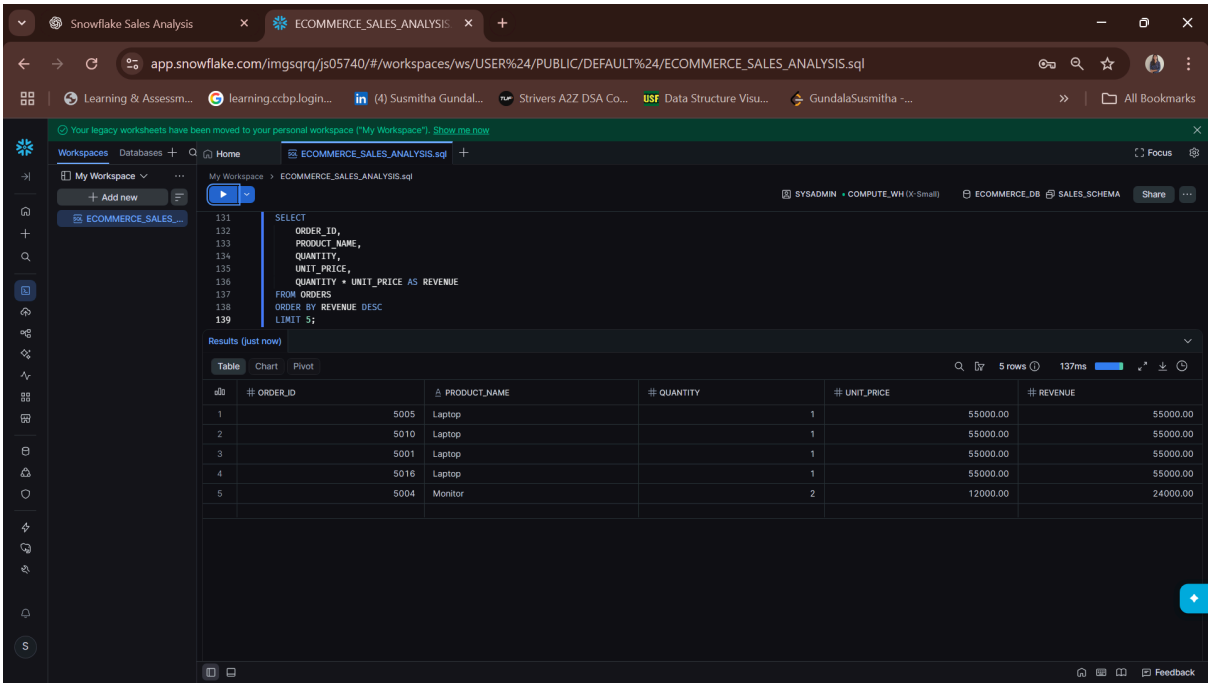
```sql
SELECT
    CUSTOMER_NAME,
    CUSTOMER_TYPE,
    CASE
        WHEN CUSTOMER_TYPE IS NULL THEN 'NULL VALUE'
        WHEN TRIM(CUSTOMER_TYPE) = '' THEN 'BLANK VALUE'
        ELSE 'VALID VALUE'
    END AS DATA_QUALITY_STATUS
FROM CUSTOMERS;
```

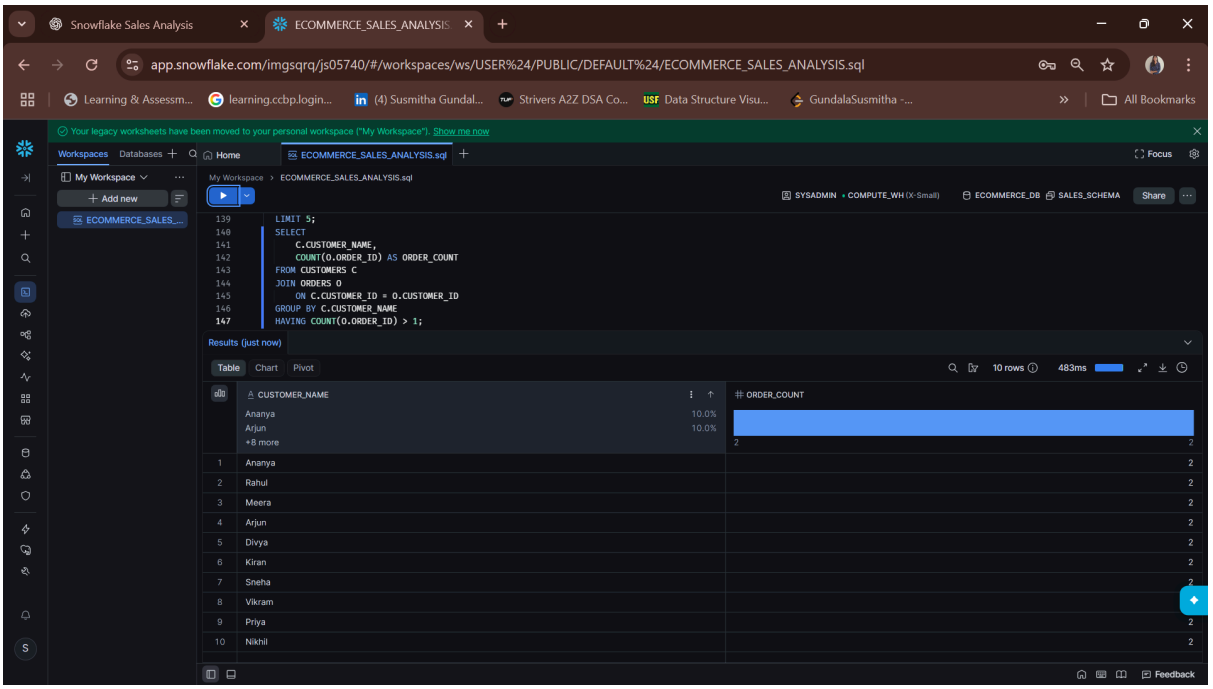



```sql
SELECT
    ORDER_ID,
    PRODUCT_NAME,
    QUANTITY,
    UNIT_PRICE,
    QUANTITY * UNIT_PRICE AS REVENUE
FROM ORDERS
ORDER BY REVENUE DESC
LIMIT 5;

SELECT
    C.CUSTOMER_NAME,
    COUNT(O.ORDER_ID) AS ORDER_COUNT
FROM CUSTOMERS C
JOIN ORDERS O
    ON C.CUSTOMER_ID = O.CUSTOMER_ID
GROUP BY C.CUSTOMER_NAME
```

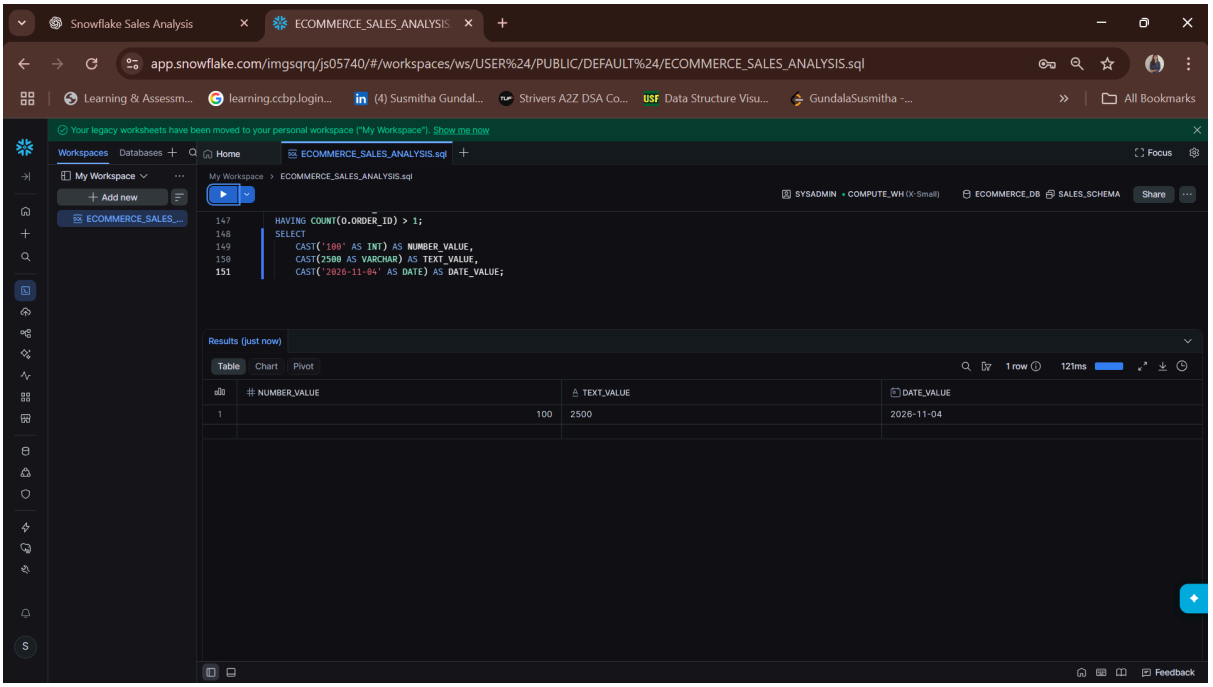
```
HAVING COUNT(O.ORDER_ID) > 1;
```





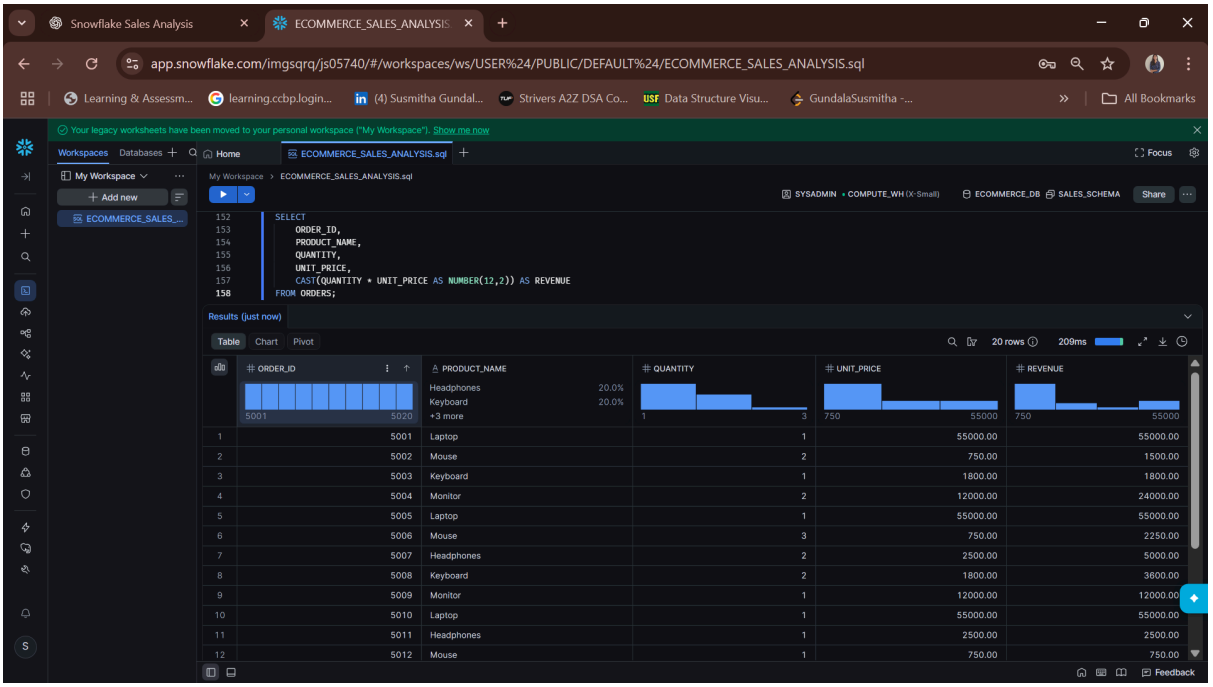
SQL

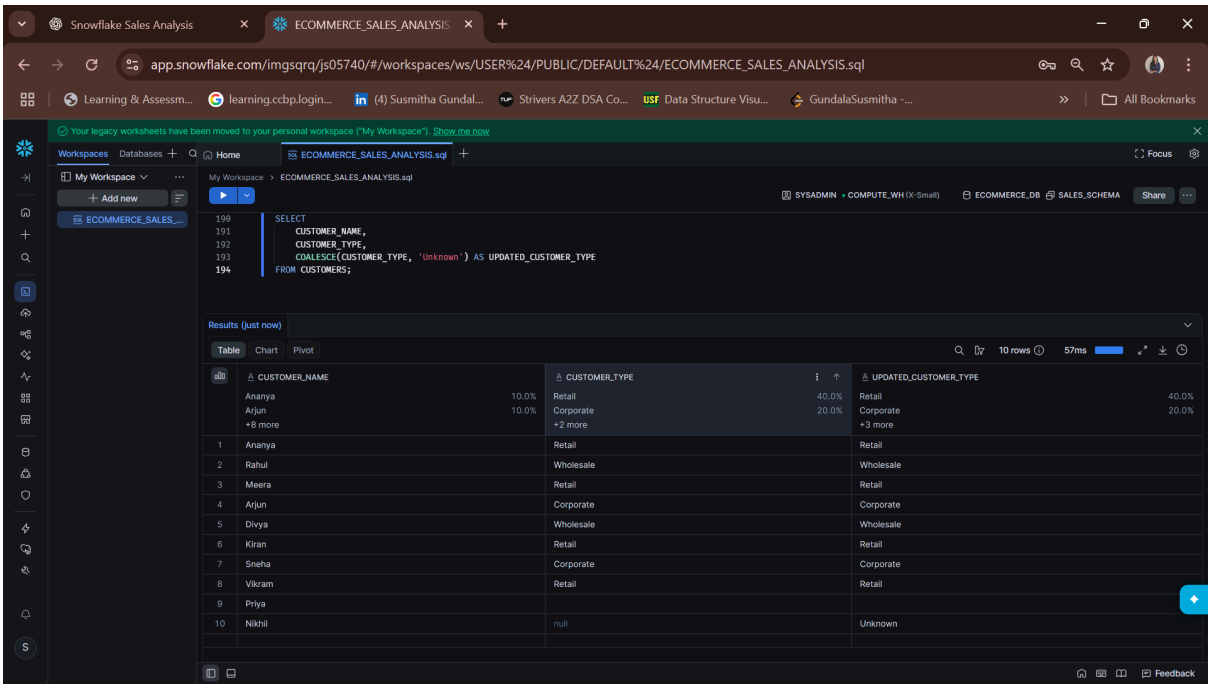
```
SELECT
    CAST('100' AS INT) AS NUMBER_VALUE,
    CAST(2500 AS VARCHAR) AS TEXT_VALUE,
    CAST('2026-11-04' AS DATE) AS DATE_VALUE;
```



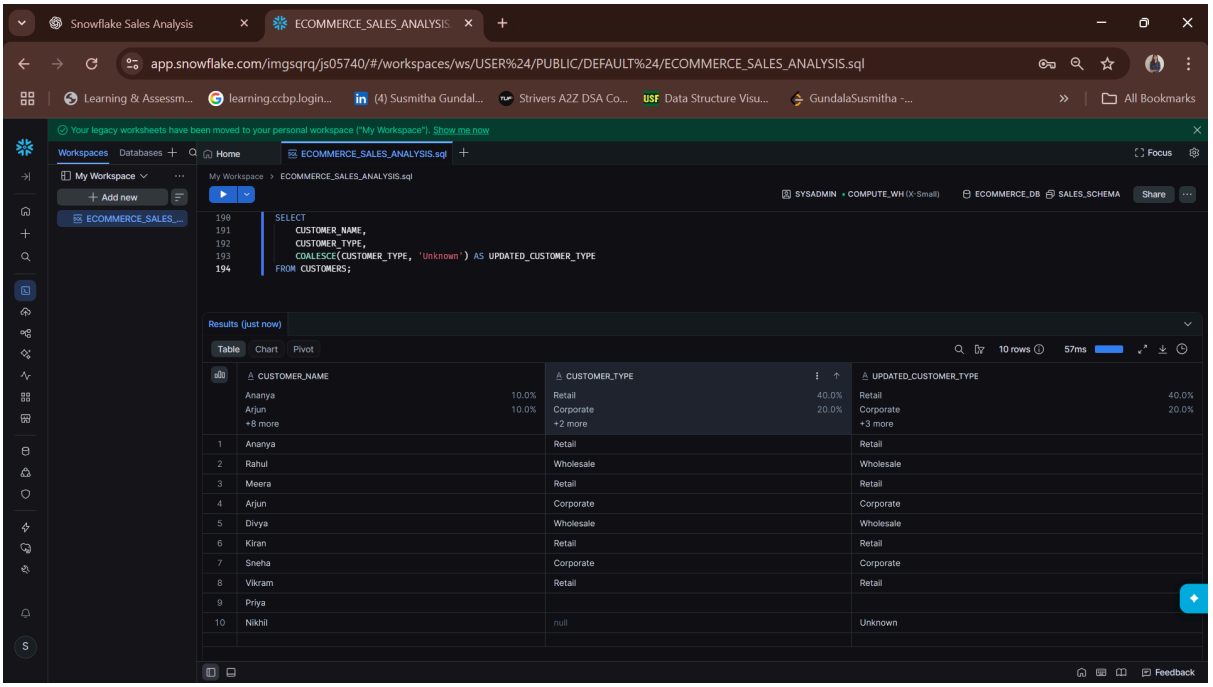

```sql
SELECT
    ORDER_ID,
    PRODUCT_NAME,
    QUANTITY,
    UNIT_PRICE,
    CAST(QUANTITY * UNIT_PRICE AS NUMBER(12,2)) AS REVENUE
FROM ORDERS;

SELECT
    CUSTOMER_NAME,
    LOCATION,
    UPPER(LOCATION) AS LOCATION_UPPERCASE
FROM CUSTOMERS;
```






```sql
SELECT
    CUSTOMER_NAME,
    CUSTOMER_TYPE,
    COALESCE(CUSTOMER_TYPE, 'Unknown') AS UPDATED_CUSTOMER_TYPE
FROM CUSTOMERS;
```


Snowflake Sales Analysis
ECOMMERCE_SALES_ANALYSIS
app.snowflake.com/imgsqrq/js05740/#/workspaces/ws/USER%24/PUBLIC/DEFAULT%24/ECOMMERCE_SALES_ANALYSIS.sql
Your legacy worksheets have been moved to your personal workspace ("My Workspace"). Show me now
Workspaces
Databases
Home
ECOMMERCE_SALES_ANALYSIS.sql
Focus
My Workspace
Add new
ECOMMERCE_SALES_...
My Workspace › ECOMMERCE_SALES_ANALYSIS.sql
SYSADMIN • COMPUTE_WH (X-Small)
ECOMMERCE_DB SALES_SCHEMA
Share
190 SELECT
191 CUSTOMER_NAME,
192 CUSTOMER_TYPE,
193 COALESCE(CUSTOMER_TYPE, 'Unknown') AS UPDATED_CUSTOMER_TYPE
194 FROM CUSTOMERS;
Results (just now)
Table
Chart
Pivot
10 rows
57ms
CUSTOMER_NAME
CUSTOMER_TYPE
UPDATED_CUSTOMER_TYPE
Ananya 10.0%
Arjun 10.0%
+8 more
Retail 40.0%
Corporate 20.0%
+2 more
Retail 40.0%
Corporate 20.0%
+3 more
1 Ananya Retail Retail
2 Rahul Wholesale Wholesale
3 Meera Retail Retail
4 Arjun Corporate Corporate
5 Divya Wholesale Wholesale
6 Kiran Retail Retail
7 Sneha Corporate Corporate
8 Vikram Retail Retail
9 Priya
10 Nikhil null Unknown
Feedback